

The sharp a.e.-existence threshold for matrix Riesz potentials

A full answer to the question in arXiv:math/0409100

Automated mathematical-discovery candidate (lane 11)

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Status. Candidate full proof; current verdict **likely valid**. In the nontrivial range of the source question, the sufficient exponent is improved to a strictly larger one and shown sharp by matching examples.

1 Source question and answer

Let $\mathfrak{M}_{n,m}$ be the real $n \times m$ matrices, $n \geq m > 1$, with $|y|_m = \det(y'y)^{1/2}$. For the orders considered in the source, the matrix Riesz potential is

$$I^\alpha f(x) = \int_{\mathfrak{M}_{n,m}} f(x-y) d\mu_\alpha(y).$$

If $a = \operatorname{Re} \alpha > m - 1$, absolute convergence is governed, up to a harmless constant, by $d\nu_a(y) = |y|_m^{a-n} dy$. If $\alpha = k$ is one of the positive integral Wallach orders, $1 \leq k \leq \min(m-1, n-m)$, μ_k is the positive rank- k measure displayed in formula (2.15) of the source.

Theorem 2.5 of [1] proves a.e. absolute convergence for every $f \in L^p$ when

$$1 \leq p < \frac{n}{a+m-1},$$

and asks whether this restriction is necessary for $m > 1$:

$$(2.16) \quad c_k = 2^{-km} \pi^{-km/2} \Gamma_m\left(\frac{n-k}{2}\right) / \Gamma_m\left(\frac{n}{2}\right).$$

Owing to (2.12), for $\alpha = 0$, μ_α is the usual delta function, and we set $I^0 f = f$. Note that the sets of α in both lines of (2.15) may overlap. In this case we have two different representations of (μ_α, ψ) .

One can use (2.14) as a definition of $I^\alpha f$, $\alpha \in \mathbf{W}_{n,m}$, for arbitrary locally integrable function provided the integral $f * \mu_\alpha$ converges absolutely.

Theorem 2.5 ([Ru6], Section 5.3). *Let $f \in L^p(\mathfrak{M}_{n,m})$, $1 \leq p < n/(Re \alpha + m - 1)$.*

(i) If $Re \alpha > m - 1$, then

$$(2.17) \quad \left| \int_{\mathfrak{M}_{n,m}} \exp(-\text{tr}(x'x)) (I^\alpha f)(x) dx \right| \leq c_1 \|f\|_p.$$

(ii) If $\alpha = k$, $k = 1, 2, \dots, n - m$, then

$$(2.18) \quad \int_{\mathfrak{M}_{n,m}} \frac{|(I^k f)(x)|}{|I_m + x'x|^{\lambda/2}} dx \leq c_2 \|f\|_p,$$

provided

$$\lambda > k + \max\left(m - 1, \frac{n + m - 1}{p'}\right), \quad \frac{1}{p} + \frac{1}{p'} = 1.$$

This statement shows that for $\alpha \in \mathbf{W}_{n,m}$ and $f \in L^p$, the definition (2.14) is meaningful, i.e., $(I^\alpha f)(x)$ is finite for almost all $x \in \mathfrak{M}_{n,m}$ provided $1 \leq p < n/(Re \alpha + m - 1)$. The last equality agrees with the classical one $1 \leq p < n/Re \alpha$ for $m = 1$ [St2] which is sharp. We do not know whether the restriction $p < n/(Re \alpha + m - 1)$ is necessary if $m > 1$.

2.3. Radon transforms on the space of matrices. The main references for this subsection are [OR1], [OR2], [Pe]. We fix positive

It is not necessary. Write

$$D = n + m - 1, \quad A = a + m - 1, \quad p_c = \frac{D}{A}.$$

The meaningful range of the question is $0 < a < n - m + 1$, so $1 < p_c < \infty$.

Theorem 1 (Sharp threshold). *Let $n \geq m > 1$ and let $\alpha \neq 0$ be either*

1. *a positive integral Wallach order $k \leq \min(m - 1, n - m)$, with $a = k$, or*
2. *an order with $m - 1 < a = Re \alpha < n - m + 1$ for which the density representation above is defined.*

Then the following are sharp.

1. *If $1 \leq p < p_c = (n + m - 1)/(a + m - 1)$, then the defining integral for $I^\alpha f$ converges absolutely for almost every x for every $f \in L^p(\mathfrak{M}_{n,m})$.*
2. *If $p_c \leq p \leq \infty$, there is a nonnegative $f \in L^p(\mathfrak{M}_{n,m})$ for which the absolute potential is infinite at every point of a fixed ball.*

For $\alpha = 0$, of course, $I^0 f = f$, so there is no restriction on p .

Since $m > 1$,

$$\frac{n+m-1}{a+m-1} > \frac{n}{a+m-1}.$$

Thus Theorem 1 both answers the source question negatively and identifies the exact replacement.

2 The smoothed-kernel lemma

Let $s_1(x) \geq \dots \geq s_m(x)$ be the singular values of x and put $d = n - a$. Let $g(x) = e^{-\text{tr}(x'x)}$. The analytic input is the following standard matrix-beta estimate, included with its reduction for auditability.

Lemma 1 (Gaussian smoothing). *For every order in Theorem 1, with total variation used in the nonreal case,*

$$(|\mu_\alpha| * g)(x) \leq C \prod_{j=1}^m (1 + s_j(x)^2)^{-d/2}. \quad (1)$$

Consequently $|\mu_\alpha| * g \in L^q(\mathfrak{M}_{n,m})$ whenever

$$dq > n + m - 1. \quad (2)$$

Proof. First suppose $m - 1 < a < n - m + 1$. The heat-kernel representation of the matrix Riesz measure (equivalently, formula (5.8) in [2]) and the change of variables $r = t^{-1}$ give, with $S = x'x/4$,

$$(\nu_a * g)(x) = C \int_{0 < r < I_m} e^{-\text{tr}(Sr)} |r|^{d/2-(m+1)/2} |I_m - r|^{a/2-(m+1)/2} dr. \quad (3)$$

Both beta parameters exceed $(m-1)/2$. Diagonalize S and successively use Schur-complement coordinates in the matrix beta integral. At each stage the only unbounded eigenvalue-dependent factor is the scalar beta Laplace integral

$$\int_0^1 e^{-\lambda u} u^{d/2-1} (1-u)^b du \leq C_b (1+\lambda)^{-d/2}, \quad b > -1.$$

The remaining Schur-complement beta integral has the same form in one lower dimension. Induction on m therefore yields

$$\int_{0 < r < I_m} e^{-\text{tr}(Sr)} d\beta_{d/2, a/2}(r) \leq C \prod_{j=1}^m (1 + \lambda_j(S))^{-d/2},$$

which is (1).

For an integral order $a = k$, insert the rank- k representation (2.15) of [1] and first perform the Gaussian integral over $\mathfrak{M}_{k,m}$. This gives

$$(\mu_k * g)(x) = C \int_{V_{n,k}} \exp\{-\text{tr}[x'(I - vv')x]\} dv. \quad (4)$$

The same Schur-complement induction, now on the Stiefel chart, gives (1) with $d = n - k$. Concretely, when one singular direction of x has size s , the set of k -frames within angle $O(s^{-1})$ of that direction has codimension $n - k$ and contributes $O(s^{-(n-k)})$; conditioning on it leaves the identical estimate for the orthogonal complement. Iteration over the singular directions gives the product in (1). This also covers the boundary (singular-beta) cases of (3).

It remains to check integrability. Polar decomposition on $\mathfrak{M}_{n,m}$ reduces the q th power of the right side of (1) to

$$C \int_{r>0} |r|^{n/2-(m+1)/2} |I_m + r|^{-dq/2} dr. \quad (5)$$

The matrix beta integral (5) is finite exactly when $dq > n + m - 1$. The necessity can already be seen by letting only its largest eigenvalue tend to infinity. This proves (2). $\square$

3 Sufficiency up to the sharp exponent

It is enough to consider $f \geq 0$. Tonelli's theorem and translation invariance give

$$\int_{\mathfrak{M}_{n,m}} g(x)(|\mu_\alpha| * f)(x) dx = \int_{\mathfrak{M}_{n,m}} f(z)(|\mu_\alpha| * g)(z) dz. \quad (6)$$

For $p = 1$, Lemma 1 makes the right side at most $C\|f\|_1$. For $1 < p < p_c$, the inequality $p < p_c$ is equivalent to

$$dp' > n + m - 1.$$

Lemma 1 and Hölder's inequality again make (6) finite. Since g is strictly positive everywhere, the absolute potential is finite for almost every x . This proves Theorem 1(1).

4 Sharp counterexamples

Set

$$E_R = \{y \in \mathfrak{M}_{n,m} : R < s_1(y) < 2R, s_2(y) < 2\}, \quad R \geq 10.$$

A tubular parametrization of the rank-one stratum gives

$$cR^D \leq |E_R| \leq CR^D, \quad D = n + m - 1. \quad (7)$$

Indeed a nonzero rank-one matrix has D parameters (including radius), and the remaining normal directions are confined to a fixed-size tube.

Take $R_j = 4^j$, so the E_{R_j} are disjoint, and put $f = \sum_j a_j \mathbf{1}_{E_{R_j}}$. Fix a sufficiently small ball B . If $x \in B$ and $y \in E_R$, Weyl's singular-value inequality gives $s_1(x - y) \leq 3R$ and $s_i(x - y) \leq 3$ for $i \geq 2$. In the density case, therefore,

$$|x - y|_m^{a-n} \geq cR^{-d}.$$

Together with (7), the j th block contributes at least

$$ca_j R_j^{D-d} = ca_j R_j^{a+m-1} = ca_j R_j^A \quad (8)$$

to the absolute potential at every $x \in B$.

For an integral order $a = k$, use formula (2.15) of [1]. Restrict the inner $\mathfrak{M}_{k,m}$ integral to matrices with largest singular value in $[6R/5, 9R/5]$ and second singular value below 1. Their measure is comparable to $R^{k+m-1} = R^A$, uniformly in the orthogonal parameter, and adding $x \in B$ places them in E_R . Hence (8) holds in the integral-order case as well.

If $p = p_c$, choose $a_j = R_j^{-A}/j$. Then

$$\|f\|_p^p \asymp \sum_j a_j^p R_j^D = \sum_j j^{-p} < \infty, \quad (|\mu_\alpha| * f)(x) \geq c \sum_j j^{-1} = \infty$$

for $x \in B$. If $p > p_c$, take $a_j = R_j^{-D/p} j^{-2/p}$; the L^p sum is finite while the terms in (8) grow exponentially. For $p = \infty$, take $a_j = 1$. This proves Theorem 1(2).

5 Scope and novelty check

The proof concerns absolute convergence of the defining convolution, exactly the meaning used around Theorem 2.5 of the source. It handles both the ordinary density range relevant to the displayed question and every positive integral Wallach order. The zero order is the explicit identity exception.

A bounded search on 12 August 2026 covered the exact title and arXiv id, the displayed exponent, “Riesz potentials on matrix spaces”, and the two candidate thresholds. It found the source, Rubin’s expanded 2006 treatment, and later book material still repeating the old sufficient condition, but no paper proving the sharp criterion above. Novelty is plausible, not certified. Human review should focus on the Schur-complement induction in Lemma 1, particularly its singular-beta/Stiefel boundary case.

References

- [1] G. Ölafsson, E. Ournycheva, and B. Rubin, *Multiscaled wavelet transforms, ridgelet transforms, and Radon transforms on the space of matrices*, arXiv:math/0409100, especially Theorem 2.5 and formulas (2.14)–(2.16).
- [2] B. Rubin, *Zeta integrals and integral geometry in the space of rectangular matrices*, arXiv:math/0406289; published as *Riesz potentials and integral geometry in the space of rectangular matrices*, Adv. Math. 205 (2006), 549–598, especially the heat-kernel representation and Section 5.3.